Unified Image Tokenization with FiLM-based Decoupling for Generation and Understanding

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Abstract

Recent advances in multimodal large language models have highlighted the importance of jointly modeling image generation and understanding, yet existing approaches struggle to reconcile their inherently conflicting requirements. Generation favors fine-grained, information-preserving representations, while understanding relies on semantically aligned abstractions, leading prior work to either decouple encoders or rely on discrete tokenization with limited expressiveness. In this work, we propose HyperTok, a unified image encoding framework that supports both tasks within a single architecture. Our key idea is to introduce a hypernetwork-based FiLM modulation mechanism, where quantized latent tokens dynamically generate task-specific conditioning parameters, enabling feature-level interaction while preserving a shared representation space. Furthermore, we adopt Finite Scalar Quantization (FSQ) to enhance token expressiveness and reduce quantization error without increasing architectural complexity. The resulting design enables effective task-aware adaptation without sacrificing representation capacity. Extensive experiments demonstrate that HyperTok achieves strong performance on both image reconstruction and multimodal understanding benchmarks, outperforming prior unified tokenizers and narrowing the gap with task-specific models. These results suggest that unified visual representation learning can be achieved without compromising either generation quality or semantic alignment.

1 Introduction

Recent advances in multimodal large language models (MLLMs) have established image generation and image understanding as two fundamental capabilities for evaluating model performance [13, 22, 42]. While these abilities are often studied in isolation [8, 15, 43, 10, 38], jointly modeling generation and understanding is crucial, as the two are inherently complementary. However, effectively integrating these two capabilities into a unified framework remains a challenge, limiting the mutual enhancement between them.

To bridge the gap between generation and understanding, a line of recent work explores dual-encoder designs [36, 4], where separate image encoders are employed to extract high-level semantic and low-level visual tokens, respectively. These heterogeneous tokens are then jointly processed within a shared transformer backbone, enabling interactions between semantic and visual information and eliminating the need to train separate models for each task. While this approach introduces a degree of integration, it comes at the cost of increased architectural and computational complexity. More

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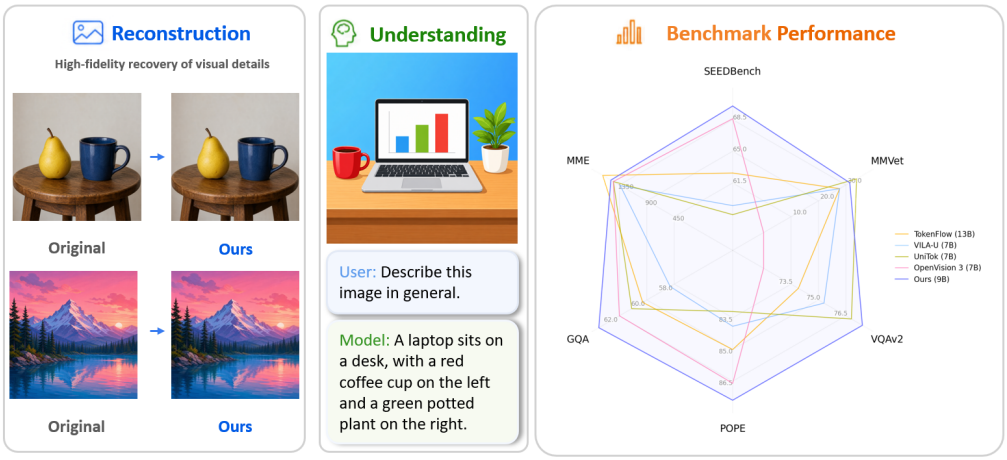


Figure 1: Our method unifies high-fidelity image reconstruction and fine-grained image understanding, while achieving strong performance across diverse multimodal benchmarks.

importantly, the separation of encoding pathways inherently limits deep feature-level fusion, hindering the model's ability to fully unify generation and understanding within a single representation space.

More recently, several works attempt to unify generation and understanding [26, 22, 45] within a single encoder. Most of these approaches rely on discrete tokens to represent images, which reduce the bitrate and naturally align with the token-based interface of large language models. However, discretization inevitably introduces quantization errors and limits representational capacity, making it difficult to accommodate the competing objectives of generation and understanding. Some methods, such as OpenVision3 [45] and TUNA [21], further incorporate pretrained models to boost performance, but this does not fundamentally resolve the optimization conflict between the two tasks.

The core challenge in unifying image generation and understanding lies in the conflicting requirements on visual representations: generation favors fine-grained, information-preserving features, while understanding relies on semantically aligned abstractions. Existing approaches either separate these signals across multiple encoders or compress them into discrete tokens, both of which lead to suboptimal trade-offs. In particular, limited token expressiveness and insufficient interaction between semantic and visual pathways hinder effective joint optimization. This suggests that a unified model must not only maintain high-capacity representations, but also enable task-aware modulation to reconcile the differing objectives.

In this work, we propose HyperTok, a unified image encoder that jointly supports generation and understanding within a single architecture. Our key idea is to introduce hypernetwork-based FiLM modulation, where quantized latent tokens dynamically generate conditioning parameters that adapt downstream modules for different tasks. This allows semantic and visual signals to interact at the feature level while preserving a shared representation space. In addition, we replace conventional vector quantization with Finite Scalar Quantization (FSQ) [23] to enhance token expressiveness and reduce quantization error, significantly improving both reconstruction fidelity and semantic performance. Together, these designs enable more effective integration of generation and understanding without increasing architectural complexity.

Contributions. Our main contributions are summarized as follows:

- We propose **HyperTok**, a unified image encoding framework that jointly supports image generation and understanding within a single architecture.
- We introduce a **hypernetwork-based FiLM modulation** mechanism, enabling task-aware feature adaptation that facilitates deeper interaction between semantic and visual representations.
- We adopt **Finite Scalar Quantization (FSQ)** to improve token expressiveness and reduce quantization error, leading to significant gains in both generation quality and semantic performance.

2 Related Work

2.1 Image Quantization Methods

Quantization is widely used for image representation, especially in generative modeling. VQ-VAE [32] introduces a learnable codebook to discretize visual features, while VQ-GAN [9] improves perceptual quality with adversarial training. Subsequent works extend this paradigm to enhance expressiveness, including multi-codebook quantization (MCQ) [22], residual vector quantization (RVQ) [16], and Finite Scalar Quantization (FSQ) [23]. However, these methods primarily focus on reconstruction and compression, often lacking the capacity to capture high-level semantic information required for unified generation and understanding.

2.2 Visual Representation for Multimodal Models

Image understanding requires representations that capture high-level semantic information beyond low-level visual details. To this end, a line of work focuses on learning semantic-aligned visual embeddings. Contrastive models such as CLIP [27] and SigLIP [44, 31], as well as self-supervised approaches like DINO [2] and DINOv2 [24], have demonstrated strong capability in learning semantically meaningful features that generalize well across downstream tasks. More recently, efforts have been made to bridge continuous representations with discrete tokenization. Methods such as BEiT v2 [25] explore discretizing semantic features derived from pretrained models, aiming to combine the strengths of representation learning and token-based modeling. However, these approaches typically treat semantic learning and generative modeling separately, lacking a unified framework that jointly supports both generation and understanding.

Existing efforts to unify generation and understanding can be broadly categorized into two paradigms.

Decoupled encoders. A line of work models visual and semantic information using separate encoders. Methods such as BAGEL [5] and Janus [36] combine generative tokenizers (e.g., VQ-VAE [32] or FLUX-VAE [15]) with pretrained semantic encoders like SigLIP [44], while RAE [46] directly leverages DINO [24] features for generation. Although these approaches benefit from strong pretrained representations, the separation of encoding pathways limits feature-level interaction and leads to fragmented representations.

Unified tokenizers. Another line of work adopts a single encoder to learn shared representations for both tasks. TokenFlow [26] introduces dual codebooks with a shared mapping, VILA-U [37] applies residual quantization on SigLIP features, and UniTok [22] employs multi-codebook quantization to improve expressiveness. More recent methods, such as TUNA [21] and OpenVision3 [45], further incorporate pretrained backbones (e.g., FLUX-VAE [15] or ViT [7]) to enhance representation quality. However, despite using a unified architecture, these methods often struggle to balance reconstruction fidelity and semantic alignment, resulting in suboptimal performance on either generation or understanding.

3 Method

3.1 Overview

We aim to learn a unified image representation that jointly supports both **generation** and **understanding**. Given an input image $x \in \mathbb{R}^{H\times W\times 3}$, our model encodes it into a set of latent tokens $\mathbf{z} = \{z_i\}_{i=1}^{T}$, which are shared across both tasks.

As illustrated in Figure 2, the latent tokens are processed by two task-specific branches: (i) a reconstruction decoder for image generation task, and (ii) a semantic projection head for understanding representation learning. To address the intrinsic conflict between generation and understanding, we introduce a **hypernetwork-based FiLM modulation** mechanism that enables **task-level decoupling within the decoder**.

Rather than relying on a shared decoding pathway that must compromise between competing objectives, our approach allows the latent tokens to dynamically modulate decoder layers via FiLM. This effectively reduces interference between tasks during optimization, encouraging the tokenizer to

preserve both fine-grained visual information and semantically meaningful abstractions, instead of collapsing to an averaged representation.

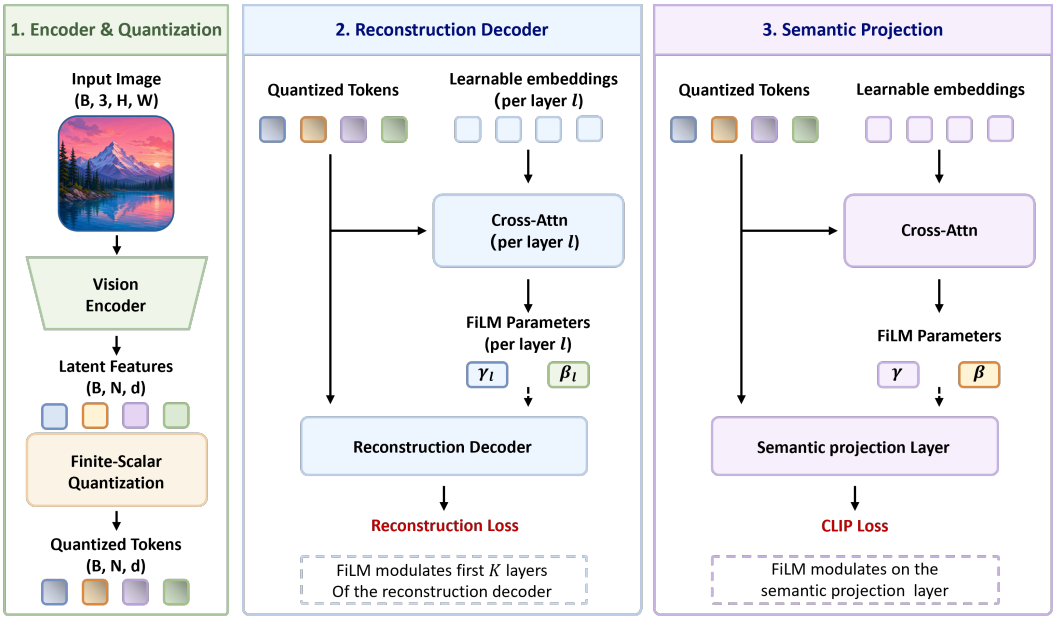


Figure 2: Overview of the HyperTok pipeline. The input image is encoded into latent features and quantized into discrete tokens, which are used for image reconstruction and CLIP-space semantic alignment. A layer-wise HyperFiLM module generates FiLM parameters via cross-attention and modulates the first K layers of both the decoder and semantic branch.

3.2 Encoder and Quantization

We adopt a **Transformer-based encoder** $\mathcal{E}_\theta$ that maps the input image to continuous patch-level latent features:

$$\mathbf{h} = \mathcal{E}_\theta(x), \quad \mathbf{h} \in \mathbb{R}^{N \times d}. \tag{1}$$

We adopt **Finite Scalar Quantization (FSQ)** to obtain discrete latent tokens. Unlike conventional Vector Quantization (VQ) or Multi-codebook Quantization (MCQ) [**?**], FSQ employs a **fixed grid-based codebook** instead of learning dynamic discrete embeddings. Each feature dimension is independently quantized by mapping continuous values to the nearest grid point. This design avoids codebook learning overhead while providing higher representational flexibility and reduced quantization error.

The quantization process is defined as:

$$\mathbf{z} = \mathcal{Q}(\mathbf{h}), \quad \mathbf{z} = \mathrm{Round}\left((L-1)\,\sigma(\mathbf{h})\right), \tag{2}$$

where $\sigma(\cdot)$ denotes the element-wise sigmoid function, L is the number of quantization levels per dimension, and $\mathrm{Round}(\cdot)$ maps each value to the nearest integer. This formulation performs **dimension-wise discretization**, resulting in $\mathbf{z} \in \{0, 1, \ldots, L-1\}^d$.

Since the rounding operation is non-differentiable, we adopt the straight-through estimator (STE) during backpropagation:

$$\mathbf{z} = (L-1)\sigma(\mathbf{h}) + \mathrm{sg}\left[\mathrm{Round}\left((L-1)\sigma(\mathbf{h})\right) - (L-1)\sigma(\mathbf{h})\right], \tag{3}$$

where $\mathrm{sg}[\cdot]$ denotes the stop-gradient operator.

Compared to vector quantization, FSQ avoids learning a codebook and instead performs scalar quantization on a fixed grid, leading to a simpler formulation with improved representational flexibility and reduced quantization error.

3.3 FiLM-based Decoder

To reconcile the conflicting objectives of generation and understanding, we design a unified **FiLM-based decoder framework** that enables task-specific modulation while preserving a shared representation space.

Given the quantized latent tokens $\mathbf{z}$, we introduce a hypernetwork $\mathcal{H}(\cdot)$ that generates feature-wise affine parameters:

$$(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \mathcal{H}(\mathbf{z}), \tag{4}$$

which are used to modulate intermediate features via FiLM:

$$\mathrm{FiLM}(f) = \boldsymbol{\gamma} \odot f + \boldsymbol{\beta}, \tag{5}$$

where f denotes the intermediate activations.

Instead of enforcing a single decoding pathway to accommodate multiple tasks, we apply FiLM modulation to the first few layers of each decoder, with the number of modulated layers being task-specific. This effectively decouples optimization in the decoder, reducing interference between generation and understanding, while enabling the shared tokenizer to capture both fine-grained visual details and high-level semantics.

3.4 Task-specific Instantiations

Reconstruction Decoder. For image generation, we adopt an attention-based decoder $\mathcal{D}_\phi$:

$$\hat{x} = \mathcal{D}_\phi(\mathbf{z}), \tag{6}$$

where FiLM parameters are generated via a task-specific hypernetwork $\mathcal{H}_{\text{rec}}$ and applied to the first K layers. This allows the decoder to adaptively emphasize low-level visual fidelity during reconstruction.

Semantic Projection Head. For semantic learning, we employ a projection head $\mathcal{P}_\psi$ composed of linear layers:

$$\mathbf{y} = \mathcal{P}_\psi(\mathbf{z}), \tag{7}$$

with FiLM parameters generated by $\mathcal{H}_{\text{sem}}$. The modulation enables the projection head to focus on semantically relevant features, improving alignment with language representations.

3.5 Training Objectives

Our model is trained with a multi-objective loss that jointly optimizes reconstruction and semantic alignment:

$$\mathcal{L} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{sem}}\mathcal{L}_{\text{sem}}. \tag{8}$$

Reconstruction Loss. The reconstruction objective consists of a pixel-level loss, a perceptual loss, and an adversarial loss:

$$\mathcal{L}_{\text{rec}} = \|x - \hat{x}\|_1 + \lambda_p \mathcal{L}_{\text{LPIPS}}(x, \hat{x}) \quad + \lambda_{\text{adv}}\mathcal{L}_{\text{GAN}}. \tag{9}$$

Semantic Loss. We adopt a CLIP-style contrastive loss over a batch of N image-text pairs $\{(x_i, t_i)\}_{i=1}^N$. Let $\mathbf{y}_i$ and $\mathbf{t}_i$ denote the normalized image and text embeddings, respectively. The loss is defined as:

$$\begin{aligned} \mathcal{L}_{\text{sem}} = &- \frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(\mathrm{sim}(\mathbf{y}_i, \mathbf{t}_i)/\tau)}{\sum_{j=1}^{N} \exp(\mathrm{sim}(\mathbf{y}_i, \mathbf{t}_j)/\tau)} \\ &- \frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(\mathrm{sim}(\mathbf{y}_i, \mathbf{t}_i)/\tau)}{\sum_{j=1}^{N} \exp(\mathrm{sim}(\mathbf{y}_j, \mathbf{t}_i)/\tau)}. \end{aligned} \tag{10}$$

Here $\mathrm{sim}(\cdot, \cdot)$ denotes cosine similarity and τ is a temperature parameter.

3.6 Discussion

Our design enables a unified representation that balances the competing requirements of generation and understanding. FSQ improves token expressiveness, while FiLM-based modulation allows task-specific adaptation without breaking the shared representation space.

4 Experiment

4.1 Implemented Details

We adopt ViTamin-L/16 [3] as our encoder, a hybrid architecture that combines convolutional layers and Transformers, which benefits both reconstruction and understanding tasks. In the main experiments, we set the FSQ levels to [-3,3] with a latent dimensionality of 128. For the reconstruction decoder, FiLM modulation is applied to the first five layers. For the understanding branch, we construct a projection head composed of two linear layers, with FiLM applied to the first layer. The FiLM module at each layer is implemented as a lightweight cross-attention head. For contrastive learning, we employ a pretrained DINOv3-S [28] model as the discriminator.

We train our model for one full epoch on the DataComp-1B [12] dataset, which contains 1.4B image-text pairs. Due to broken links, we were able to successfully download half of the dataset (approximately 700M) image–text pairs in practice. As this scale is insufficient to train a CLIP model from scratch, we initialize the CLIP text encoder with pretrained weights and fine-tune it during training. The entire training process uses the Muon optimizer with a global batch size of 16K. The learning rate is set to 1×10^{-3} for the tokenizer, and 2×10^{-4} for both the discriminator and the CLIP text encoder.

4.2 Evaluation Setting

We conduct three groups of main experiments. The first evaluates the reconstruction capability of the learned tokenizer, which reflects whether it preserves complete and high-fidelity visual information. The second group evaluates the effectiveness of our tokenizer in downstream class-conditional image generation. Following the training protocol of RAE [46], we train a class-conditional generator on ImageNet. The third group assesses performance on downstream image understanding tasks. We adopt the LLaVA-Next [20] training framework and use Qwen-3.5 9B [30] as the base model to train a vision-language model.

4.3 Reconstruction performance

As shown in Tab. 4, our method achieves superior reconstruction performance across nearly all metrics compared to previous works. In particular, we obtain state-of-the-art results on PSNR, SSIM, and LPIPS, which directly measure per-image reconstruction quality.

Although our method yields slightly worse rFID on ImageNet compared to UniTok, it significantly outperforms the baseline on COCO. Specifically, our method reduces rFID from 3.918 to 0.981 on COCO, achieving approximately a $4\times$ improvement. Given that COCO images typically exhibit more complex and diverse visual content, this result indicates that our approach is more robust to distribution shifts, maintaining strong reconstruction quality even under increased scene complexity, whereas the baseline suffers from substantial degradation.

We also show a visual comparision between RAE [46], UniTok [22] and HyperTok. As can be seen, RAE fails to accurately reconstruct the general color and details of the original image. Although UniTok recovers the overall structure, it still struggles to preserve high-frequency details, such as faces and text. In contrast, our algorithm achieves near-perfect reconstruction, making it difficult for the human eye to distinguish between our output and the original image.

4.4 Understanding Performance

As shown in Tab. 4, our method achieves state-of-the-art or near state-of-the-art performance across a wide range of image understanding benchmarks among discrete tokenizers. Specifically, we obtain

Table 1: Comparison of image reconstruction performance on ImageNet and COCO. We report PSNR, SSIM, LPIPS, and rFID. Continuous-token-based models are shown in gray for reference, while results are compared among discrete tokenizers. Our method (HyperTok) achieves the best overall performance, demonstrating strong reconstruction fidelity.

Models	ImageNet [6]				COCO [19]			
	PSNR↑	SSIM↑	LPIPs↓	rFID↓	PSNR↑	SSIM↑	LPIPs↓	rFID↓
Generation-oriented Tokenizer								
SD3-VAE [8]	31.29	0.886	0.059	0.201	31.18	0.894	0.056	1.671
Cosmos [1]	25.07	0.700	0.167	0.959	24.74	0.711	0.165	5.063
FLUX-VAE [15]	32.86	0.917	0.044	0.176	32.73	0.923	0.041	1.343
Wan2.1-VAE [33]	31.34	0.886	0.058	0.945	31.19	0.895	0.055	3.449
Unified Tokenizer								
RAE [46]	17.44	0.403	0.324	1.060	16.98	0.394	0.345	10.119
OmniTokenizer [34]	24.69	0.771	0.138	1.411	24.31	0.779	0.137	6.292
Vila-U [37]	22.24	0.612	0.228	4.231	21.89	0.620	0.227	10.997
UniTok [22]	25.34	0.742	0.132	**0.362**	24.95	0.750	0.131	3.918
HyperTok	**30.40**	**0.896**	**0.059**	0.591	**29.61**	**0.883**	**0.063**	**0.981**

Table 2: Comparison of multimodal understanding performance on standard benchmarks, including VQAv2, GQA, POPE, MME, MM-Vet, and SeedBench. Continuous-token-based models are shown in gray for reference. Among discrete tokenizers, our method (HyperTok) achieves state-of-the-art performance on multiple benchmarks, highlighting its strong semantic representation capability.

Models	Size	Token Type	Res.	VQAv2 val	GQA test-dev	POPE acc.	MME perc.	MM-Vet acc.	SeedBench image
Emu3 [35]	8B	Continuous	512	77.3	60.3	85.2	–	37.2	68.2
Janus [36]	7B	Continuous	384	77.3	59.1	87.0	1338	34.3	63.7
Janus-Pro [4]	7B	Continuous	384	–	62.0	87.4	1567	50.0	72.1
Mogao [18]	7B	Continuous	512	–	60.9	88.9	1592	–	74.6
Show-o2 [40]	7B	Continuous	1024	–	63.1	–	1621	–	69.8
Tuna [21]	7B	Continuous	512	–	63.9	–	1642	–	74.7
Show-o [39]	1.3B	Discrete	256	59.3	48.7	73.8	948	–	–
TokenFlow [26]	13B	Discrete	256	73.9	60.3	85.0	**1623**	27.7	62.6
Chameleon [29]	34B	Discrete	512	69.6	–	–	–	–	–
Vila-U [37]	7B	Discrete	256	75.3	58.3	83.9	1336	27.7	59.0
UniTok [22]	7B	Discrete	256	76.8	61.1	83.2	1448	**33.9**	–
OpenVision 3 [45]	7B	Discrete	256	–	62.0	86.6	1449	–	68.6
HyperTok	9B	Discrete	256	**77.4**	**63.5**	**87.4**	1489	31.4	**70.0**

the best results on VQAv2 [13], GQA [14], POPE [17], and SeedBench [41], while ranking second on MME [11] and MM-Vet [42].

Notably, our method consistently outperforms prior approaches even when compared to models with significantly larger base architectures. For example, compared to TokenFlow (13B) [26] and Chameleon (34B) [29], our method (9B) achieves higher performance on VQAv2 (77.4 vs. 73.9, +3.5) and GQA (63.5 vs. 60.3, +3.2), demonstrating stronger efficiency and scalability.

Furthermore, despite operating with discrete tokens, our method matches or even surpasses continuous-token-based approaches on several benchmarks. For instance, we achieve 77.4 on VQAv2 and 63.5 on GQA, which are comparable to or better than strong continuous baselines such as Emu3 [35] (77.3 / 60.3) and Janus [36] (77.3 / 59.1). This demonstrates that the tokenizer learned by our approach is capable of capturing rich semantic information, bridging the gap between discrete representations and high-level visual understanding.

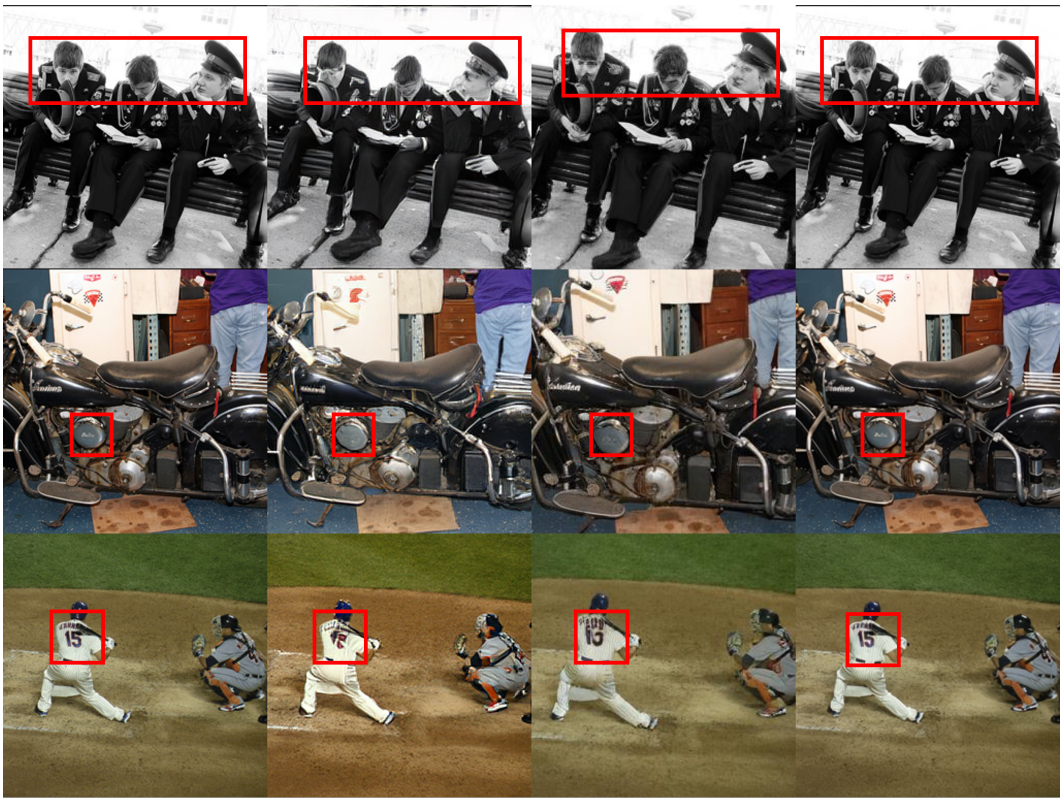


Figure 3: **Reconstruction visualization.** We present a visual comparison of image reconstruction quality among RAE, UniTok, and our method. Our method produces high-fidelity reconstructions that effectively preserve the details of the original images.

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